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Human milk metabolite extraction for NMR analysis

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Edison Lab | Breast Milk Protocol

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We use this protocol and it's working

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Disclaimer

This protocol is intended for research use only. Users should follow their institutional safety and ethical requirements when handling human samples and hazardous chemicals.

Abstract

This is a protocol for a modified dual-phase extraction of human milk samples, originally described by Andreas et al. (2015).

Also see:

Citation

Andreas NJ, Hyde MJ, Gomez-Romero M, Lopez-Gonzalvez MA, Villaseñor A, Wijeyesekera A, Barbas C, Modi N, Holmes E, Garcia-Perez I
(2015). Multiplatform characterization of dynamic changes in breast milk during lactation. Electrophoresis.

https://doi.org/10.1002/elps.201500011 LINK

Guidelines

- Use glass tubes compatible with organic solvents.
- Label all tubes.
- Randomize samples.
- Calibrate pipettes.
- Keep samples on ice during the extraction procedure.

Materials

1. Samples
- Human breast milk samples

2. Chemicals and reagents

- Chloroform/methanol solution (2:1, v/v)
- Purified water
- D_2O phosphate buffer (pH 7.4) containing 3-(trimethylsilyl)propionic acid (TSP) as an internal standard
- Deuterated chloroform ($CDCl_3$) containing tetramethylsilane (TMS) as an internal standard
- Ice

3. Consumables

- Pipette tips suitable for the required volume ranges
- 5 mL capped glass tubes
- 1.5 mL capped glass tubes
- 5 mm SampleJet NMR tubes (Bruker)
- Sample labels

4. Equipment

- Calibrated micropipettes covering volumes from 100 to 1,000 µL
- Refrigerated centrifuge capable of reaching 13,480 × g at 4 °C
- Vortex mixer
- Vacuum concentrator (SpeedVac) suitable for aqueous and organic solvents
- Ice bucket or cooling rack

Safety warnings

- Handle all breast milk samples as potentially biohazardous material and wear appropriate personal protective equipment.
- Chloroform and methanol are hazardous and volatile solvents. Handle them in a chemical fume hood and dispose of solvent waste according to institutional safety procedures.

Ethics statement

This protocol was applied to human breast milk samples collected from 20 donors recruited at Hospital Gineco Obstétrico Isidro Ayora in Quito, Ecuador. All procedures were approved by the Ethics Committee of Pontificia Universidad Católica del Ecuador (PUCE; approval code MB-06-2021, approved on November 4, 2021) and were conducted in accordance with the Declaration of Helsinki. Participant recruitment and sample collection followed the procedures established in the approved study protocol.
Prior approval from an appropriate institutional ethics committee must be obtained before using this protocol to collect or analyze human samples in other studies.

Before start

Confirm that all required reagents, materials, and equipment are available.

Day 0 - BEFORE STARTING

1 Thaw frozen breast milk samples On ice or at 4 °C prior to sample preparation and metabolite extraction.

Day 1 -DUAL PHASE EXTRACTION

2 Briefly vortex the original sample, then transfer 400 µL of the original human breast milk sample to a new 5-mL glass tube.

3 Add 2 mL of chloroform/methanol (2:1, v/v) to each human breast milk sample and keep the mixture on ice.

4 Mix the mixture thoroughly by vortexing for 15 seconds.

5 Add 600 µL of purified water to the sample mixture.

6 Vortex the mixture for 15 seconds.

7 Centrifuge the sample for 10 min at 13480 x g at t 4 °C .

Note

This extraction method produces two distinct phases.

8 Carefully transfer 800 µL each phase into a separate new 1.5-mL glass tube. Avoid disturbance or collection of the interphase.

Note

Avoid disturbing or collecting material from the interphase, as this may cause cross-contamination between the polar and non-polar extracts.

9 Evaporate the polar and non-polar phases in separate tubes using a vacuum concentrator (SpeedVac).

10 Once dry, keep the samples at -80 °C until analysis.

Day 2 – Reconstitution of Dried Extracts for 1H NMR Analysis

11 Add 700 µL of a D2O phosphate buffer to each dried polar extract and mix until completely reconstituted.

Day 2 - ORGANIC PHASE RECONSTITUTION

12 Add 700 µL of deuterated chloroform to the ORGANIC phase dried extract.

13 Transfer the reconstituted organic phase into 5 mm NMR tubes for 1H NMR spectroscopy.

Protocol references

Andreas, N. J., Hyde, M. J., Gomez-Romero, M., Lopez-Gonzalvez, M. A., Villaseñor, A., Wijeyesekera, A., ... & Garcia-Perez, I. (2015). Multiplatform characterization of dynamic changes in breast milk during lactation. Electrophoresis, 36(18), 2269-2285.

Citations

Andreas NJ, Hyde MJ, Gomez-Romero M, Lopez-Gonzalvez MA, Villaseñor A, Wijeyesekera A, Barbas C, Modi N, Holmes E, Garcia-Perez I. Multiplatform characterization of dynamic changes in breast milk during lactation. **https://doi.org/10.1002/elps.201500011**

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